

CS-1180 Lab Problem 2: Grade Calculator - IO

PURPOSE: To test the ability to read input from the user, format print statements, and review data types

DIRECTIONS: For this problem, you will be using the code from the Lab Problem 1: Grade Calculator as a starting point. You should modify your code to prompt the user to enter their grade for the ZyBooks, Attendance, Lab, and Midterm exam categories, as well as their desired total grade for the course. Essentially, instead of hard-coding the grades, you should be asking the user to provide them as console input and then using those values in your calculations. (Note: You may still hard-code the weights. They will not be changing any time soon, regardless of what a student may wish.) Again, make sure to use the most appropriate data type for each value.

For each grade that the user enters, print out a line which repeats the entered grade, assignment name, and its associated weight. When printing out the details, you **MUST** use the variables for the grades, names, and weights you created in the first lab problem rather than hard-coding the output.

Your output for each grade category should look similar to the output below for zyBooks (depending on the values the user gives, of course):

```
You have a 97.30% grade for zyBooks, which is 10% of your final grade.
```

Next, print the student's calculated current average grade. It should look similar to the output below:

```
Your grade so far is 90.51%.
```

Next, prompt the student for their desired grade in the class. Using this value, finally calculate and print the student's target final exam grade. Again, you **must** use the variables you created to hold these values in the first lab problem when printing the results rather than hard-coding the output. It should look similar to the output below:

```
To get a 90.00% in the class, you need to get a 87.94% on your final exam.
```

As shown in the examples given above, **round all displayed grades to TWO decimal places**. When rounding numbers for display, it is best to not override the stored values themselves with less precise rounded versions. Instead, use `System.out.printf()` or `String.format()` to round only the displayed values.

...*HOW* do you use these methods to round to two decimal places? Hint: Use the token `%.2f`.

..."But if format tokens use a % sign, how do I get % signs themselves to display correctly for the grades and weights?" Good question ... ;-)

Note: You do not need to verify that what the user enters is the correct data type or in any particular range. It is acceptable (for now!) if your program crashes when bogus input is given.

EXAMPLE: Running your program should look like this. (It is fine if your input is on the same line as the prompt for that input.)

```
What is your grade for zyBooks?
> 97.3
You have a 97.30% grade for zyBooks, which is 10% of your final grade.
What is your grade for attendance?
> 100
You have a 100.00% grade for attendance, which is 5% of your final grade.
What is your grade for lab?
> 89.72
You have a 89.72% grade for lab, which is 42% of your final grade.
What is your grade for the quizzes?
> 86.24
You have a 86.24% grade for the quizzes, which is 10% of your final grade.
What is your grade for the midterm?
> 87.5
You have a 87.50% grade for the midterm, which is 13% of your final grade.
Your grade so far is 90.51%.
What is your desired grade for the class?
> 90
To get a 90.00% for the class, you need to get a 87.94% for your final exam.
```

RUBRIC:

(3 pts) Base Functionality

- [1] All print statements round correctly
- [1] All print statements use variables appropriately (no hard-coding)
- [1] User is prompted for all the required data and program uses that data for the calculations

(1 pts) Style

- [1] The code is meaningfully commented

IMPORTANT NOTES:

- Submissions that do not compile will receive a zero
- Submissions with improperly cited AI will receive a zero and an academic integrity violation
- Submissions that are partially or fully copied from another submission will receive a zero and an academic integrity violation